Identifying Physical Activity Intensity Windows Associated with Joint Space Width Using Accelerometer Data

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Background & Previous Presentation Recap

Data Description

the Osteoarthritis Initiative (OAI)

Large, longitudinal public health study

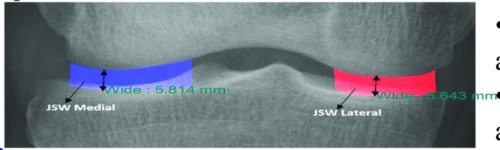
Data collected in the United States

Wearable accelerometer measurements

Minute-level physical activity counts

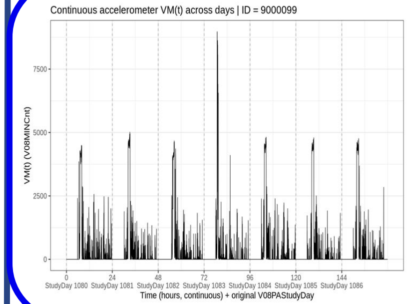
Multiple days per participant

Medial Minimum Joint Space Width (Outcome): measured in millimeters (mm), represents the minimum distance between the femur and tibia within the knee joint.



- A **larger** JSW: healthier cartilage and less structural degeneration.
- A **smaller** JSW: cartilage loss and narrowing of the joint space.

Conduct OTC

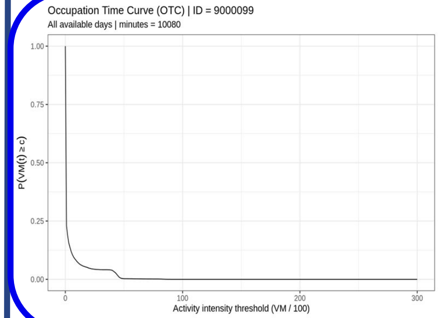


Vector Magnitude time series

VM(t): minute-level activity count at time t.

Raw data: single time axis

This step: time re-indexing



Occupation Time Curve (OTC):

$$OTC(c) = P\{VM(t) \geq c\}$$

The OTC intentionally ignores the temporal ordering of VM(t).

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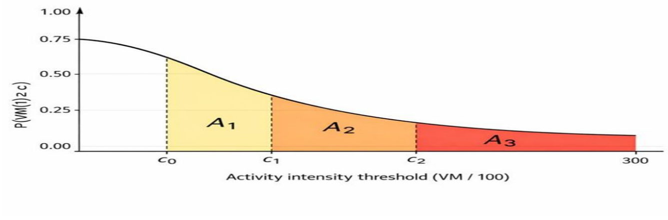
Background & Previous Presentation Recap

OTC→AUC



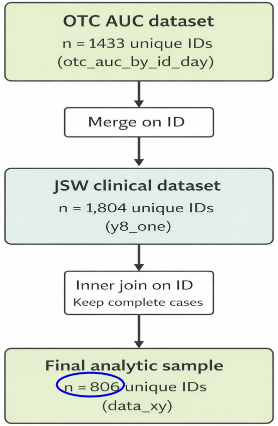
We partition the activity intensity axis into intervals:
$0 = c_0 < c_1 < \cdots < c_J$,

and define, for each interval,

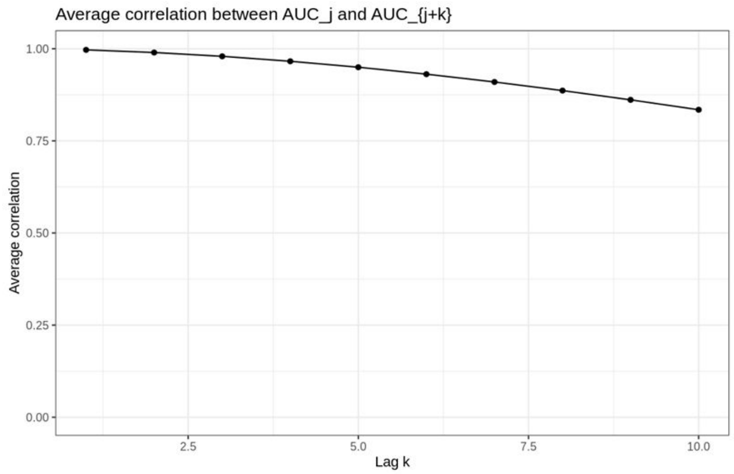
$$A_j = \int_{c_{j-1}}^{c_j} OTC(c)\, dc,$$

$j = 1, \ldots, J$.

Add Outcome



Why this method?



Strong correlation leads to applying fused lasso or standard regression unsuitable at this stage.

Functional Data Representation

Variable	Description
Y_i	**Outcome (Medial Minimum Joint Space Width)**
Z_i	**Additional covariate (currently not select)**
$X_i = (AUC_{i1}, AUC_{i2}, \ldots, AUC_{iJ})$	**Area under the activity curve across intensity intervals (Summary of time spent)**

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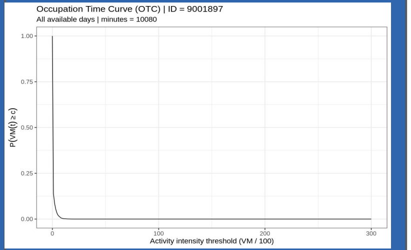
Method: Data Preparation

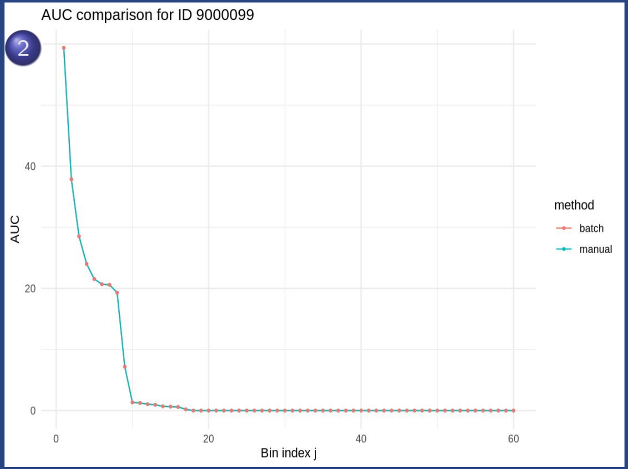
AUC Validation

1

	ID	n_minutes	AUC_1	AUC_2	AUC_3	AUC_4	AUC_5	AUC_6	AUC_7	AUC_8	AUC_9	AUC_10	AUC_11	AUC_12	AUC_13
	<int>	<int>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	9000099	10080	59.4	37.8	28.5	24.0	21.5	20.7	20.6	19.3	7.19	1.34	1.24	1.04	0.942
2	9001695	10080	73.6	39.4	23.8	14.5	10.1	7.59	5.26	3.27	2.48	1.64	1.04	0.744	0.546
3	9001897	10080	11.9	1.49	0.347	0	0	0	0	0	0	0	0	0	0
4	9002116	12960	35.0	14.9	6.21	3.16	1.97	0.926	0.617	0.270	0.231	0.0772	0	0	0
5	9002817	10080	40.0	17.8	10.3	7.94	4.81	2.28	0.744	0.0992	0	0	0	0	0
6	9003126	10080	64.6	22.9	5.36	0.595	0.0496	0	0	0	0	0	0	0	0



2




```
> print(head(check_batch, 20))
    j AUC_manual  AUC_batch diff
1   1 59.3750000 59.3750000    0
2   2 37.8472222 37.8472222    0
3   3 28.5218254 28.5218254    0
4   4 24.0079365 24.0079365    0
5   5 21.5277778 21.5277778    0
6   6 20.6845238 20.6845238    0
7   7 20.5853175 20.5853175    0
8   8 19.2956349 19.2956349    0
9   9  7.1924603  7.1924603    0
10 10  1.3392857  1.3392857    0
11 11  1.2400794  1.2400794    0
12 12  1.0416667  1.0416667    0
13 13  0.9424603  0.9424603    0
14 14  0.6944444  0.6944444    0
15 15  0.6448413  0.6448413    0
16 16  0.5952381  0.5952381    0
17 17  0.1984127  0.1984127    0
18 18  0.0000000  0.0000000    0
19 19  0.0000000  0.0000000    0
20 20  0.0000000  0.0000000    0
```

AUCs are validated in two ways:

(1) consistent with OTC structure

(2) identical between manual and batch computations

Standardization

All AUC features and the outcome are standardized (mean 0, variance 1).

- **Avoid scale dominance**

Early AUC bins are larger magnitudes, and without this step they dominate the fit.

- **Prevent biased cutpoint selection**

Scale imbalance can pull the estimated cutpoint toward early bins.

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Method: Optimization-Based Functional Modeling

Functional Model Establishment

Ideal Model: $$Y = \int \mathrm{activity}(c) \times \mathrm{effect}(c)\, dc$$

This model directly captures how the effect varies with activity intensity.

Since the observed activity data are discrete rather than continuous,

Alternative Model: $$Y_i = \sum_{j=1}^{J} X_{ij}\theta_j + \gamma Z_i + \epsilon_i$$

Build a discrete approximation of a functional regression.

Due to high correlation across AUCs and high dimension,

Set: $$\theta_j = \beta_k \quad \text{if } j \in \text{segment } k$$

This stabilizes and enables interpretation of intensity ranges.

Substitute the piecewise-constant assumption into the original model,

$$\sum_{j \in \text{segment } k} X_{ij}\theta_j = \beta_k \sum_{j \in \text{segment } k} X_{ij}$$

Final Model: $$Y_i = \sum_{k=1}^{K} \beta_k \left(\sum_{j \in \text{segment } k} X_{ij} \right) + \gamma Z_i + \epsilon_i$$

Convert pointwise effects into segment-level effects.

Benefits of the Proposed Method

- Dimensionality reduction (J → K)
- Improved stability by reducing multicollinearity
- Enhanced interpretability via segment-level effects

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Method: Optimization-Based Functional Modeling

Optimization

Optimization formulation:

$$\min \sum_i \left(Y_i - \sum_{k=1}^{K} \beta_k \left(\sum_{j \in segment\, k} X_{ij} \right) - \gamma Z_i \right)^2$$

minimize the Residual Sum of Squares (RSS)

For each selected value of K,
we estimate the corresponding segment specific coefficients under the segmentation constraints.

Model Selection

$$BIC = n \log(RSS/n) + p \log(n)$$

Select the optimal number of segments K using BIC.

n: subject size (804)
p: parameter size: (betas(K)+gamma(1)+cutpoint(K-1))

This balances model fit and model complexity:
A larger K allows more flexibility;
A smaller K leads to a simpler and more interpretable model.

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Result & Discussion: Current Achievement

Model Performance

Parameters	J = 60				J = 100				J = 200			
	K=1	K=2	K=3	K=4	K=1	K=2	K=3	K=4	K=1	K=2	K=3	K=4
β_1	0.0036	0.0171	0.0346	0.0344	0.0022	0.0105	0.0183	0.0090	0.0011	0.0052	0.0086	0.0044
β_2	—	-0.0002	0.0072	0.0077	—	-0.0001	0.0042	-0.2306	—	-0.00006	0.0021	-0.5324
β_3	—	—	-0.0023	-0.0366	—	—	-0.0013	0.2204	—	—	-0.0007	0.5254
β_4	—	—	—	0.0012	—	—	—	-0.0009	—	—	—	-0.0004
c_1 (jump)	—	11	3	3	—	18	6	28	—	37	13	58
c_2 (jump)	—	—	23	25	—	—	39	29	—	—	78	59
c_3 (jump)	—	—	—	27	—	—	—	30	—	—	—	60
γ	0	0	0	0	0	0	0	0	0	0	0	0
RSS	796.96	783.78	781.80	781.01	796.89	783.58	781.68	779.52	796.81	783.48	781.63	778.79
BIC	6.31	**6.28**	17.62	30.20	6.23	**6.07**	17.51	28.65	6.15	**5.97**	17.45	27.90

J	c_1 (jump)	c_1/J
60	11	0.183
100	18	0.180
200	37	0.185

- A **two-segment** model: consistent
- The cutpoint: approximately **18%** of the intensity range.

- **Below** this cutpoint: activity has a **positively** associated with the outcome.
- **Above** this cutpoint: the association becomes **slightly negative (negligible)**.

Result Interpretion

- The association between activity intensity and joint health is mainly driven by **lower intensity activity**.
- Engaging in **light to moderate** activity is more strongly associated with maintaining joint space, which reflects healthier cartilage and less structural degeneration.
- Increasing activity intensity **beyond** the identified cutpoint **does not** appear to provide additional benefit for joint health.

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Result & Discussion: Current Achievement

Permutation Test

Null hypothesis: The two-segment model is unnecessary: $\beta_2 = 0$.

$\Delta BIC_{obs} = BIC(K=1) - BIC(K=2)$ how K=2 improves the model over K=1 for observed data.

shuffle JSW while keeping AUC fixed, creating a dataset with no association between activity and the outcome in each permutation

$\Delta BIC^{(b)}_{perm} = BIC(K=1) - BIC(K=2)$ how K=1-to-2 improves the model on the b-th permuted dataset.

$\Delta BIC^{(1)}_{perm}, \ldots, \Delta BIC^{(B)}_{perm}$ Repeating this process B times produces a set of ΔBIC values that form the null distribution.

$$p = \frac{1 + \sum_{b=1}^{B} I\left(\Delta BIC^{(b)}_{perm} \geq \Delta BIC_{obs}\right)}{B+1}$$

Indicator function:
how often the permuted ΔBIC exceeds the observed ΔBIC.

The p-value is computed as the proportion of permuted datasets where the improvement in model fit (ΔBIC) is at least as large as that observed in the original data.

Test Result

Permuted number	P-value
B = 50	0.0196
B = 80	0.0123
B = 100	0.0099

- Improvement of K=2 over K=1 is **statistically significant**.
- The null hypothesis is evidently **rejected**.
- The permutation test **supports** the **two-segment** model structure.

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Result & Discussion: Future Arrangement

Limitations

- The suspicious minutes are not considered at this stage
- Need adjustment for confounders
- Data-driven segmentation may vary across samples

Next Steps

- Handle suspicious activity patterns (non-wear, outliers)
- Apply confounders (age, sex, BMI, baseline joint health)
- Compare with alternative modeling approaches (e.g., models used by other group members)

Reference Page

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Thank you for watching

Welcome to ask questions